slaves find their way into any reproductives at all, it will be into the reproductives that emerge from the original nest from which they were kidnapped. The slave workers will, if anything, be busy breaking the wrong code! Therefore, queens of a slave-making species can get away with changing their code freely, without there being any danger that genes for breaking the code will be propagated into the next generation.

The upshot of this involved argument is that we should expect in slave-making species that the ratio of investment in reproductives of the two sexes should approach 1:1 rather than 3:1. For once, the queen will have it all her own way. This is just what Trivers and Hare found, although they only looked at two slave-making species.

I must stress that I have told the story in an idealized way. Real life is not so neat and tidy. For instance, the most familiar social insect species of all, the honey bee, seems to do entirely the 'wrong' thing. There is a large surplus of investment in males over queens—something that does not appear to make sense from either the workers' or the mother queen's point of view. Hamilton has offered a possible solution to this puzzle. He points out that when a queen bee leaves the hive she goes with a large swarm of attendant workers, who help her to start a new colony. These workers are lost to the parent hive, and the cost of making them must be reckoned as part of the cost of reproduction: for every queen who leaves, many *extra* workers have to be made. Investment in these extra workers should be counted as part of the investment in reproductive females. The extra workers should be weighed in the balance against the males when the sex ratio is computed. So this was not a serious difficulty for the theory after all.

A more awkward spanner in the elegant works of the theory is the fact that, in some species, the young queen on her mating flight mates with several males instead of one. This means that the average relatedness among her daughters is less than $\frac{3}{4}$, and may even approach $\frac{1}{4}$ in extreme cases. It is tempting, though probably not very logical, to regard this as a cunning blow struck by queens against workers! Incidentally, this might seem to suggest that workers should chaperone a queen on her mating flight, to prevent her from mating more than once. But this would in no way help the workers' own genes—only the genes of the coming generation of workers. There is no trade-union spirit